

Опорные функции, сумма Минковского, полярность

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It is convenient to encode a convex body by lines rather than by points: the support function turns the geometry of Minkowski sums into ordinary arithmetic of functions, while polarity interchanges “wide” and “narrow”. This is the language in which mixed volumes are stated later.

Definition 1. For a nonempty compact convex $K \subseteq \mathbb{R}^n$ the **support function** is $h_K(u) = \max_{x \in K} \langle u, x \rangle$, $u \in \mathbb{R}^n$. The **width** of K in direction u ($|u| = 1$) is $h_K(u) + h_K(-u)$.

Definition 2. For K with 0 in its interior the **gauge** (Minkowski functional) is $\|x\|_K = \inf\{t > 0 : x \in tK\}$, and the **polar body** is $K^\circ = \{y : \langle x, y \rangle \leq 1 \ \forall x \in K\}$.

Theorem 1 (Minkowski). $K \mapsto h_K$ is a bijection between nonempty compact convex sets and sublinear functions ($h(u+v) \leq h(u) + h(v)$, $h(\lambda u) = \lambda h(u)$ for $\lambda \geq 0$); moreover $h_{K+L} = h_K + h_L$ and $K \subseteq L \iff h_K \leq h_L$.

Theorem 2 (bipolar theorem). If K is closed convex and $0 \in K$, then $(K^\circ)^\circ = K$. If K is a symmetric convex body, then $\|\cdot\|_{K^\circ}$ is the norm dual to $\|\cdot\|_K$.

Theorem 3 (Blaschke). The set of convex bodies in $\mathbb{R}^n$ contained in a fixed ball is compact in the Hausdorff metric.

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1. Prove Theorem 1: the identities $h_{K+L} = h_K + h_L$, $h_{\lambda K} = \lambda h_K$ and the criterion $K \subseteq L \iff h_K \leq h_L$; then show that every sublinear function is the support function of exactly one compact convex set.
 2. (VJIMC 2001, cat. I) Let A, B, C be nonempty sets in $\mathbb{R}^n$, with A bounded and C closed and convex. Prove that $A + B \subseteq A + C$ implies

$B \subseteq C$. (Hint: first the convex compact case through support functions, then the general one.)

3. Prove the bipolar theorem, and compute K° for the cube $[-1, 1]^n$, for the cross-polytope $\{\sum |x_i| \leq 1\}$ and for the ball. Prove also $(\text{conv}(K \cup L))^\circ = K^\circ \cap L^\circ$.
4. (ICMC 2021–22) A set of points has *point symmetry* if a reflection in some point maps the set to itself. Let P be a solid convex polyhedron in $\mathbb{R}^3$ whose orthogonal projection onto every plane has point symmetry. Prove that P itself has point symmetry. (Hint: look at the width and at the function $h_P(u) + h_P(-u)$.)
5. (ΦKH, vechu 2025) Let E be an infinite family of translates of an ellipse e in the plane, and suppose that every line parallel to a fixed line r meets at least one member of E . Prove that there are infinitely many triples of members of E met by one common line. (Hint: reduce affinely to unit discs; “a line meets three discs means that the three centres lie in a strip of width 2.”)
6. Prove that $K \mapsto h_K$ is an isometry from the Hausdorff metric to the uniform metric on the sphere, deduce Blaschke’s selection theorem from Arzelà–Ascoli, and use it to show that among all planar convex bodies of a given perimeter there is one of maximal area.